

Standard form

Astronomers, biologists, engineers, physicists and many others encounter quantities whose measures involve very small or very large numbers. For example, the distance of the Earth from the Sun is approximately 144 000 000 000 metres and the distance that light will travel in 1 year is 5 870 000 000 000 metres.

It is sometimes tedious to write or work with such numbers. This difficulty is overcome by writing such numbers in standard form.

$$\text{E.g. } 144\,000\,000\,000 = 1.44 \times 10^{11}$$

$$5\,870\,000\,000\,000 = 5.87 \times 10^{12}$$

If a quantity is written as the product of a power of 10 and a number that is greater than or equal to 1 and less than 10, then the quantity is said to be expressed in **standard form** (or **scientific notation**).

$$\text{For example, } 658 = 6.58 \times 10^2$$

Note:

- We have expressed 658 as a product of 6.58 and a power of 10. Clearly, 6.58 is between 1 and 10. So the standard form of 658 is 6.58×10^2 .
- The digit 6 in 658 is in the hundreds (10^2) place

In general:

In converting a number to standard form, the place value of the first digit of the number is the power of ten.

$$\text{E.g. } 56 = 5.6 \times 10^1$$

$$564 = 5.64 \times 10^2$$

$$5648 = 5.648 \times 10^3$$

$$0.05648 = 5.648 \times 10^{-2}$$

Note:

5 is in the hundredths (10^{-2}) place.